

Project 1: Exploratory of Historical Stock Data Analysis

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Technical Report

Notebook link -

http://localhost:8888/notebooks/Stock_Market_Data_Part_1.ipynb

My overall approach

I wanted to explore a large stock-market dataset in a clear, step-by-step way, so that even someone with no technical background could follow what I did and why. I always moved in the same order: first understand the data, then clean it, then explore it, then visualize it, then investigate anything unusual, and finally summarize. I never jumped to conclusions before understanding the data first.

Step 1 — Loading and understanding the data

I used the pandas library to load both files (the prices file and the company file) into tables. I then looked at the first few rows, the number of rows and columns, and the data types. I did this so I knew exactly what I was working with — like reading the label before opening a box.

Step 2 — Cleaning and checking quality

I checked for missing values, duplicate rows, and impossible values (for example, a negative price, or a "high" price that was lower than the "low" price — which can't happen in real trading). I followed one important rule: **investigate first, don't delete something just because it looks unusual**. I only removed data that was clearly broken, like a junk row that had header words instead of real company data, and I wrote down every decision I made.

Step 3 — Fixing the date

The date column was stored as plain text, so the computer didn't understand it as a real date. I converted it to a proper date format. This mattered because you can't do any time-based analysis — like sorting by year or plotting a trend — until the computer knows the column is actually a date.

Step 4 — Why I used three different describe methods

describe() is a function that gives all the key statistics at once (the average, the middle value, the spread, the smallest and the largest). I used it in three ways, each for a reason:

- A **plain describe()** for a quick first look.
- A version that showed **only the number columns** and left out the date, so the table was cleaner and easier to read.
- A **display setting** that showed the huge volume numbers as full numbers instead of confusing scientific notation (like 1.85e+09). This just made the results readable for a normal person.

So it was one tool used three ways to make the numbers as clear as possible.

Step 5 — Why I looked at the top 10 across different companies

When I first listed the 10 highest closing prices, they all belonged to a single stock (KTOS). That wasn't very useful, because it only told me about one company. So instead of using that simple code, I used a small technique that showed the highest price for **10 different companies**. This let me confirm that high prices weren't a one-time glitch — several real companies (including Apple) genuinely trade at high prices. It gave me more confidence that the data was real, not full of errors.

Step 6 — Why I chose Plotly Express for the charts

I chose Plotly Express instead of a basic charting tool because it creates **interactive charts**. I can hover over any point to see the exact value, zoom into busy areas, and inspect outliers directly. This made it much easier to explain the data and to spot unusual points, especially in a stock's price trend over time.

Step 7 — Investigating unusual values the right way

Whenever I found extreme values, I followed the process **Detect → Investigate → Document → Decide**, never **Detect → Delete**. For example, before deciding whether the very high KTOS prices were errors, I checked that stock's own history and compared its closing price with its adjusted closing price. Only after investigating did I decide to keep them, because they were real.

Step 8 — Looking at one stock in detail

Finally, I picked one stock (KTOS) and plotted its closing price over time. This showed a clear boom-and-bust story: a big spike in early 2000, then a long decline. Looking at one stock closely helped me connect all the earlier numbers to a real-world example.

Result

By following this simple, repeatable flow, I turned a large and slightly messy dataset into a clear picture. The data is real and mostly clean, most stocks are low-priced with a few high-priced exceptions, and the overall data is strongly right-skewed with a small number of extreme values. Each step built on the one before it, so the conclusions are based on evidence, not guesses.